

Def  $\sum_{k=1}^{\infty} a_k$ ,  $R_m = \sum_{k=m+1}^{\infty} a_k$ ,  $m \in \mathbb{N} \cup \{0\}$

коз.  $m$  - Тыи остатком  $\mu$  пере

$$\underline{\text{Zam}} \quad \sum_{k=1}^{\infty} a_k = S_m + R_m$$

$\mathcal{A} \quad \sum_{k=1}^{\infty} a_k = \text{cs.} \Leftrightarrow \forall R_m = \text{cs.} :$

$$\sum_{k=1}^{\infty} a_k = S_m + R_m$$

$$\Delta \quad ] \quad n > m \Rightarrow \sum_{k=1, \dots, s_n}^n a_k = \sum_{k=1}^m a_k + \sum_{k=m+1}^n a_k \quad (\Leftarrow)$$

$$\Leftarrow \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \lim_{n \rightarrow \infty} \sum_{k=1}^m a_k + \lim_{n \rightarrow \infty} \sum_{k=m+1}^n a_k$$

$$\Leftrightarrow R_m \in \mathbb{R} : S_{\substack{\in \mathbb{R}}} = S_m_{\substack{\in \mathbb{R}}} + R_m \quad \Delta$$

$$\sum_{k=1}^{\infty} a_k - c_n \Leftrightarrow \lim_{m \rightarrow \infty} R_m = 0$$

$$\Delta(H) \supset \sum_{k=1}^{\infty} a_k - \infty \stackrel{11}{\Rightarrow} S = S_m + R_m$$

$$\Rightarrow S = \lim_{m \rightarrow \infty} S_m + \lim_{m \rightarrow \infty} R_m \Rightarrow \lim_{m \rightarrow \infty} R_m = 0$$

⑨  $\exists \lim_{m \rightarrow \infty} R_m = 0 \Rightarrow \forall m \ R_m \in \mathbb{R} \Rightarrow$

$$\exists \lim_{n \rightarrow \infty} R_1 \in \mathbb{R} \Rightarrow \exists \lim_{n \rightarrow \infty} R_n \in \mathbb{R} \quad \forall n \in \mathbb{N}$$

$$\uparrow \Rightarrow \sum_{k=1}^{\infty} a_k - \text{con. } \Delta$$

u ]  $\sum_{k=1}^{\infty} a_k, \sum_{k=1}^{\infty} b_k - \text{cs.} \Rightarrow \forall \alpha, \beta \in \mathbb{R}$   
 $\sum_{k=1}^{\infty} \alpha a_k + \beta b_k - \text{cs.} \quad u$

$$\sum_{k=1}^{\infty} (\alpha a_k + \beta b_k) = \alpha \sum_{k=1}^{\infty} a_k + \beta \sum_{k=1}^{\infty} b_k$$

$$\Delta ] S^A = \sum_{k=1}^{\infty} a_k \quad S_n^A = \sum_{k=1}^n a_k$$

$$S^B = \sum_{k=1}^{\infty} b_k \quad S_n^B = \sum_{k=1}^n b_k$$

$$\Rightarrow S_n = \sum_{k=1}^n (\alpha a_k + \beta b_k) = \alpha \sum_{k=1}^n a_k + \beta \sum_{k=1}^n b_k =$$

$$= \alpha S_n^A + \beta S_n^B, \Rightarrow \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} (\alpha S_n^A + \beta S_n^B) =$$

$$= \alpha \lim_{n \rightarrow \infty} S_n^A + \beta \lim_{n \rightarrow \infty} S_n^B = \alpha S^A + \beta S^B \in \mathbb{R}$$

$$\Rightarrow \exists S = \lim_{n \rightarrow \infty} S_n = \alpha S^A + \beta S^B \quad \Delta$$

$$\textcircled{11} ] a_k \leq b_k, \quad \sum_{k=1}^{\infty} a_k = S^A \in \overline{\mathbb{R}} \\ \sum_{k=1}^{\infty} b_k = S^B \in \overline{\mathbb{R}} \Rightarrow \sum_{k=1}^{\infty} a_k \leq \sum_{k=1}^{\infty} b_k$$

$$\Delta ] S_n^A = \sum_{k=1}^n a_k \quad S_n^B = \sum_{k=1}^n b_k$$

$$] a_k \leq b_k \Rightarrow S_n^A \leq S_n^B \xRightarrow{\lim}$$

$$\lim_{n \rightarrow \infty} S_n^A \leq \lim_{n \rightarrow \infty} S_n^B \Rightarrow S^A \leq S^B \quad \Delta$$

$\parallel^A$   $\parallel^B$   
 $S^A$   $S^B$

Признаки сравнения

$\textcircled{17}$  Критерий  $\text{сн} - \text{ти}$   $\sum_{k=1}^{\infty} a_k$ ,  $a_k \geq 0$

$$] a_k \geq 0 \Rightarrow \begin{cases} \{ S_n \} - \text{возрастает} \\ \sum_{k=1}^{\infty} a_k = \sup_{n \in \mathbb{N}} S_n \end{cases}$$

$$\Rightarrow \sum_{k=1}^{\infty} a_k - \text{сн} \Leftrightarrow \{ S_n \} - \text{суп.}$$

$$\Delta \textcircled{4} \quad \exists a_k \geq 0 \Rightarrow \underline{S_{n+1} = S_n + a_{n+1} \geq S_n}$$

$$\Rightarrow \{S_n\} \uparrow \stackrel{\text{TB-oe}}{\Rightarrow} \{S_n\} - \text{ср.}, \text{ если } \{S_n\} - \text{ср.}$$

$$\textcircled{5} \quad \exists \begin{cases} \{S_n\} - \text{ср.} \\ \{S_n\} - \text{ср.} \end{cases} \stackrel{\text{TB-oe}}{\Rightarrow} \{S_n\} - \text{ср.} \Rightarrow \sum_{k=1}^{\infty} a_k \text{ ср.}$$

$\textcircled{T}$  *типичные свойства.*

$$\exists 0 \leq a_k \leq b_k \Rightarrow$$

$$\textcircled{1} \quad \text{Если } \sum b_k - \text{ср.} \Rightarrow \sum a_k - \text{ср.}$$

$$\textcircled{2} \quad \text{Если } \sum a_k - p | \text{ср.} \Rightarrow \sum b_k - p | \text{ср.}$$

$$\textcircled{3} \quad \text{Если } a_k \sim b_k \text{ при } k \rightarrow \infty \Rightarrow$$

$$\begin{cases} \sum a_k & \text{ср. или } p | \text{ср.} \\ \sum b_k & \text{одновременно} \end{cases}$$

$$\Delta \textcircled{1} \quad \exists S_n^A = \sum_{k=1}^n a_k \quad \begin{cases} S_n^B = \sum_{k=1}^n b_k \\ S^B = \sum_{k=1}^{\infty} b_k \in \mathbb{R} \end{cases}$$

$$\text{т. к. } a_k \leq b_k \Rightarrow S_n^A \leq S_n^B \leq S^B < +\infty$$

$$\Rightarrow \{S_n^A\} - \text{ср. и } \uparrow \Rightarrow \exists \lim_{n \rightarrow \infty} S_n^A = S^A$$

$$2). \quad \exists \begin{cases} \sum a_k - p | \text{ср.} \\ \sum b_k - \text{ср.} \end{cases} \stackrel{\textcircled{1} \uparrow}{\Rightarrow} \sum a_k - \text{ср.} - \text{нр-ые}$$

$$\Rightarrow \sum b_k - p | \text{ср.}$$

$$3) \quad \exists a_k \sim b_k, \Rightarrow \begin{cases} a_k = d_k b_k, \\ \lim_{k \rightarrow \infty} d_k = 1 \end{cases} \Rightarrow$$



$$\Rightarrow \forall \varepsilon > 0 \exists N_\varepsilon(\varepsilon) \in \mathbb{N}: \forall k \geq N_\varepsilon \Rightarrow$$

$$|a_k - 1| < \varepsilon \Rightarrow 1 - \varepsilon < a_k < 1 + \varepsilon \quad / \cdot b_k$$

$$(1 - \varepsilon) b_k < a_k b_k < (1 + \varepsilon) b_k \Rightarrow$$

$$(1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$$

$$\textcircled{1} \exists \sum b_k - \text{cs.} \Rightarrow \sum (1 + \varepsilon) b_k - \text{cs.} \stackrel{\textcircled{1} \uparrow}{\Rightarrow} \sum a_k - \text{cs.}$$

$$\textcircled{2} \exists \sum b_k - \text{p.lcs.} \Rightarrow \sum (1 - \varepsilon) b_k - \text{p.lcs.} \stackrel{\textcircled{2} \uparrow}{\Rightarrow} \sum a_k - \text{p.lcs.}$$

$$\textcircled{3} \exists \sum a_k - \text{cs.} \Rightarrow \sum (1 - \varepsilon) b_k - \text{cs.} \Rightarrow \sum b_k - \text{cs.}$$

$$\textcircled{4} \exists \sum a_k - \text{p.lcs.} \Rightarrow \sum (1 + \varepsilon) b_k - \text{p.lcs.} \Rightarrow \sum b_k - \text{p.lcs.}$$

*Tip* 1)  $\sum_{k=1}^{\infty} \frac{1}{k^k} \quad \frac{1}{k^k} = a_k > 0$

$$\frac{1}{k^k} \leq \frac{1}{2^k}, \quad k \geq 2 \quad \Rightarrow \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k - \text{cs.}$$

$$\Rightarrow \text{no 1. pr. pr. cs. u. ncs. here.}$$

$$2) \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \quad \frac{1}{\sqrt{k}} > 0$$

$$\frac{1}{\sqrt{k}} \geq \frac{1}{k}, \quad \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k} - \text{p.lcs.}$$

$$\Rightarrow \text{no 1. pr. pr. p.lcs. u. ncs. here}$$

$$3) \sum_{k=1}^{\infty} \left( e^{\arcsin \frac{1}{k}} - 1 \right)^{\alpha}$$

$$\arcsin x = x + o(x^2), \quad x \rightarrow 0$$

$$e^x - 1 = x + o(x), \quad x \rightarrow 0$$

$$e^{\arcsin \frac{1}{k}} - 1 \sim \frac{1}{k} \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} = \begin{cases} \text{cs.} & \text{if } \alpha > 1 \\ \text{p.lcs.} & \text{if } \alpha \leq 1 \end{cases}$$

$$\Rightarrow \text{ncs. here cs. if } \alpha \geq 1$$



# Т) Правило Коши

$$\exists a_k > 0, \quad \exists \lim_{k \rightarrow \infty} \sqrt[k]{a_k} = l \in [0, +\infty] \Rightarrow$$

① если  $l > 1$ , то  $\sum a_k$  — п.с.

② если  $l < 1$ , то  $\sum a_k$  — с.с.

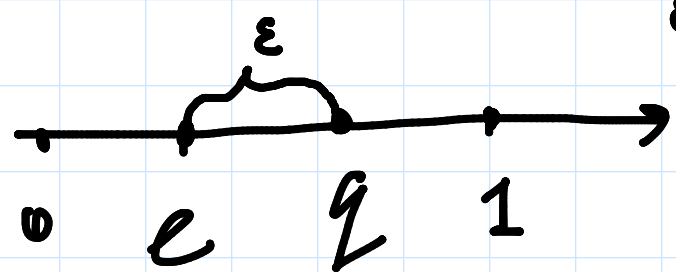
А 1)  $\exists \{a_{k_n}\} : \lim_{n \rightarrow \infty} \sqrt[k_n]{a_{k_n}} = l$

$$\exists l > 1 \Rightarrow \forall n \gg N_\varepsilon \Rightarrow \sqrt[k_n]{a_{k_n}} > 1 \Rightarrow$$

$$\Rightarrow a_{k_n} > 1 \Rightarrow \lim_{n \rightarrow \infty} a_{k_n} \neq 0 \Rightarrow \sum a_k \text{ — п.с.}$$

2)  $\exists l < 1, \exists q : l < q < 1 \Rightarrow$

$$\forall \varepsilon > 0 \exists N_\varepsilon \in \mathbb{N} : \forall k \gg N_\varepsilon \Rightarrow \left| \sqrt[k]{a_k} - l \right| < \varepsilon$$



$$\exists q - l = \varepsilon$$

$$\sqrt[k]{a_k} < q < 1$$

$$a_k < q^k < 1$$

$$\sum q^k \text{ — геом. прогр.: } q < 1 \Rightarrow \text{с.с.}$$

$$\Rightarrow \text{по 1 пр. ср. с.с. и } \sum a_k.$$

иначе  $\exists \{a_{k_n}\} : \lim_{n \rightarrow \infty} \sqrt[k_n]{a_{k_n}} \neq l$

# Т) Правило Д'Аламбера

$$\exists a_k > 0 \text{ и } \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = l \in [0, +\infty] \Rightarrow$$

① если  $l > 1 \Rightarrow \sum a_k$  — п.с.

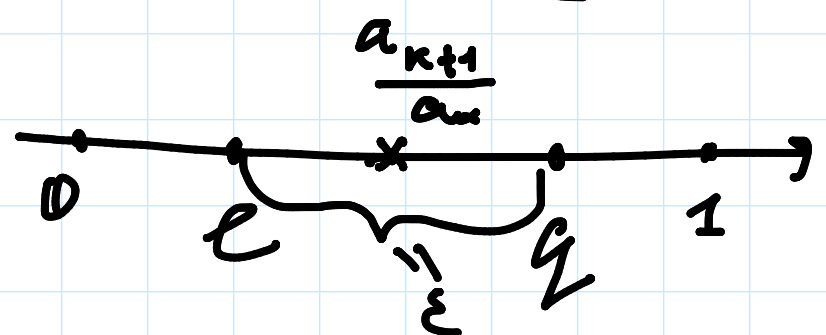
② если  $l < 1 \Rightarrow \sum a_k$  — с.с.

А 1)  $\exists l > 1 \Rightarrow \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1 \Rightarrow \forall k \gg N =$

$$\frac{a_{k+1}}{a_k} > 1 \Rightarrow a_{k+1} > a_k \Rightarrow \lim_{k \rightarrow \infty} a_k \neq 0$$

$\Rightarrow$  по дост. уч. при help  $\sum a_k - p | a_n$

2)  $\exists l < 1$   $\exists q : l < q < 1$  !

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = l \Rightarrow$$


$\forall \varepsilon > 0 \exists N_\varepsilon \in \mathbb{N} : \forall k > N_\varepsilon \Rightarrow$

$$\left| \frac{a_{k+1}}{a_k} - l \right| < \varepsilon \Rightarrow l - \varepsilon < \frac{a_{k+1}}{a_k} < l + \varepsilon$$

Достаточно  $q - l = \varepsilon \Rightarrow \frac{a_{k+1}}{a_k} < q$

$\Rightarrow a_{k+1} < q a_k, k > N_\varepsilon \Rightarrow$

$$a_{N+1} < q a_N$$

$$a_{N+2} < q a_{N+1} < q^2 a_N$$

...

$$\begin{aligned} & (a_1 + a_2 + \dots + a_N) + (a_{N+1} + \dots) \\ & a_N + (q a_N + q^2 a_N + \dots), \quad q < 1 \end{aligned}$$

начиная с  $a_{N+1} : a_{N+k} < q^k a_N$

$\sum q^k a_N = \text{сн} \Rightarrow$  по 1-й сф. сн  $\sum a_k$   $\triangle$